An Unsupervised Dehazing Network With Hybrid Prior Constraints for Hyperspectral Image

Wei He, *Senior Member, IEEE*, Mengyuan Wang, Yong Chen, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*— Haze pollution in hyperspectral images (HSIs) leads to surface information lack and image clarity degradation, which seriously affects the performance of subsequent image interpretation. Existing model-based hyperspectral haze removal methods enjoy good interpretability and generalization, but they can only process images in a specific wavelength range due to the principle limitation. Deep learning-based dehazing methods have good feature extraction capability, but the cost of acquiring sufficient training data is high in practical applications. At the same time, taking into account that HSIs have spectral low-rank structures, fully utilizing the low-rank property will facilitate the reconstruction of HSIs. In order to combine the complementary benefits of deep learning-based and physical model-based approaches, we decide to formulate HSI dehazing reconstruction as an unsupervised deep image prior (DIP) framework. Specifically, we propose an unsupervised dehazing network with hybrid prior constraints (HPC-UDN) for HSI haze removal, which effectively integrates low-rank prior, deep priors, and physical haze prior. First, the low-rank prior of hyperspectral data is characterized by matrix decomposition, where the decomposition factors are learned through two generative networks. Then, multiple spectral groups are divided based on the correlation and complementarity between spectral bands. In order to exchange information between adjacent spectral groups, a novel spectral grouping feature fusion (SGFF) module is designed, which connects neighboring spectral groups to transfer spectral and spatial features. Finally, high-quality HSI is recovered by merging the features extracted from each spectral group. Extensive simulated and real-data experiments certify the effectiveness and robustness of the presented unsupervised approach and potential applications in the GF-5 image dehazing task.**

***Index Terms*— Deep generative network, haze removal, hyperspectral image (HSI), low-rank matrix decomposition, unsupervised learning.**

Manuscript received 4 November 2023; revised 17 January 2024 and 3 March 2024; accepted 7 April 2024. Date of publication 12 April 2024; date of current version 22 April 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903501; in part by the National Natural Science Foundation of China under Grant 42271370; and in part by the Natural Science Foundation of Jiangxi, China, under Grant 20232ACB212001 and Grant 20224BAB212001. *(Corresponding author: Hongyan Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3388245

I. Introduction

HYPERSPECTRAL images (HSIs), which can display spatial features and spectral characteristics of ground objects simultaneously, have been employed in a lot of applications, such as mineral exploration [1], environmental assessment [2], and military target reconnaissance [3]. However, owing to atmospheric absorption and reflection effects, captured hyperspectral data are often contaminated by haze, causing the captured data to exhibit weakened color and texture features, spectral distortion, and low contrast. Image quality degradation seriously affects the performance of subsequent image interpretation [4], [5]. Consequently, HSI dehazing is crucial to improve image interpretation accuracy.

Different from natural images that only have three bands of red, green, and blue (RGB), remote sensing images have tens or hundreds of bands. The huge spectral difference between natural images and HSIs makes the dehazing methods developed for natural images unsuitable for hyperspectral/remote sensing image dehazing. Therein, many remote sensing image dehazing techniques have been specifically proposed. We can divide these techniques into three categories: image enhancement-based methods [6], [7], physical model-based methods [8], [9], and deep learning-based methods [10], [11]. Image enhancement-based methods aim to improve image contrast with traditional image processing. Without considering the degradation mechanism, it is difficult to accurately restore the spatial–spectral information obscured by haze. Relying on the atmospheric scattering model (ASM), physical model-based methods use certain priori assumptions to restore clear images from the imaging essence of hazy images. These model-based methods can improve the visual effect in some cases, but the hand-designed prior cannot be adapted to all haze scenes. To compensate for this deficiency, deep learning-based techniques use a host of hazy-clean image pairs to learn the mapping relationship between them. Although encouraging dehazing results have been achieved, the cost of obtaining sufficient training data is high, which cannot be applied well in practice.

Physical model-based methods enjoy good interpretability and generalization, and deep learning-based methods have good feature extraction capabilities. In order to combine the advantages of both, we hope to organically integrate physical priors and deep priors described by deep networks into one

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framework. At the same time, considering that HSIs have discriminative intrinsic spectral low-rank structures, taking full advantage of the low-rank property will facilitate HSI reconstruction. Based on the above considerations, we decide to formulate HSI dehazing reconstruction as an unsupervised deep image prior (DIP) framework, which can make the best use of hybrid benefits of deep learning-based and physical model-based techniques. Accordingly, we have proposed an unsupervised dehazing network with hybrid prior constraints (HPC-UDN). **First**, we introduce the deep low-rank model and convert the HSI dehazing problem to the extrapolation of the spectral basis and the spatial representation coefficient. **Second**, we calculate the correlation coefficient (CC) of all bands and then partition the spectral groups to fully exploit the spectral correlation. **Finally**, each spectral branch is convolved in parallel to capture haze features. To adaptively integrate information between neighboring spectral groups, we elaborately design a novel SGFF module, which transmits spectral information from longer-wavelength branches to shorter-wavelength branches to recover spectral characteristics. The proposed dehazing network not only exploits the low-rank property of HSI but also takes advantage of the deep priors inherent in deep neural networks while taking into account the physical property that haze levels vary across spectral bands. By organically combining physical priors and deep priors, additional training data is not needed for the proposed HPC-UDN. It optimizes the parameters in an unsupervised manner to recover clean HSI from the original hazy HSI. The following is a summary of the main contributions.

1) We combine the hybrid benefits of model-based and learning-based techniques, which effectively integrate low-rank prior, deep priors, and physical haze prior for HSI haze removal. First, we perform low-rank matrix decomposition on hyperspectral data, where the decomposition factors are learned by two deep generative networks. Then, the spectral grouping is performed based on the high correction and complementarity between adjacent bands. Finally, feature extraction is performed on each spectral branch to obtain a clean HSI.
2) In remote sensing images, the resistance to haze interference increases gradually with the wavelength increases, which means the available helpful features in the longer-wavelength group contribute to recovering the shorter-wavelength group. In order to exchange information between adjacent spectral groups, a novel SGFF module is proposed, which connects neighboring spectral groups to transfer spectral and spatial features.
3) In our experiments, the proposed method provides excellent results on both simulated images destroyed by different levels of fog and real-data images, demonstrating its potential application in the practical GF-5 image dehazing task.

The rest of this article is structured as follows. Section II introduces relevant works on HSI haze elimination. The detailed steps of the proposed HPC-UDN are elaborated in Section III. Section IV uses a series of simulated and real-data experiments to certify the defogging effectiveness of the presented HPC-UDN. Section V presents the conclusion and research directions for the future.

II. Related Work

A. HSI Dehazing Methods

In recent years, a great quantity of remote sensing image dehazing techniques have been proposed. We can categorize these techniques into image enhancement-based techniques, physical model-based techniques, and deep learning-based techniques. Instead of focusing on the degradation mechanism, image enhancement-based approaches aim to improve image details and visibility by means of image processing such as contrast enhancement and histogram equalization [6]. Ancuti and Ancuti [12] implemented a pixel fusion technique to improve outdoor scene visibility, where the input information is obtained from the original foggy image through a white balance and contrast-enhancing procedure. Li et al. [7] designed a visual enhancement algorithm to improve the contrast and brightness of hazy images based on the histogram distribution prior. Huang et al. [13] developed an urban scene dehazing method by combining the multiscale retinax theory and image phase consistency features. Richter [14] matched the histograms of hazy areas and clear scenes to obtain the ground reflectance data. Furthermore, homomorphic filtering [15], retinex enhancement [16], and wavelet transform [17] were also introduced for remote sensing image defogging. Image enhancement-based methods enjoy low complexity and high real-time performance, but the dehazing effect is unsatisfactory due to ignoring the physical mechanism.

Physical model-based haze removal methods employ ASMs and certain prior knowledge to estimate transmission maps and global atmospheric light (AL). Combining the dark channel prior and saturation prior, Xie et al. [9] implemented automatic transmission map estimation without manually setting parameters. By adding a translation term, Pan et al. [8] designed a deformed haze imaging model, which enables the imaging model to achieve the leap from natural images to remote sensing images. Zhu et al. [18] utilized the color attenuation prior to restoring image scene depth and thus estimated the transmittance matrix. Wang et al. [19] enabled an ovel patch-based haze removal method, which describes the AL distribution using the AL surface hypothesis rather than the global invariant plane. Makarau et al. [20] further advanced the dark object subtraction to estimate hazy thickness maps for multispectral data. Shen et al. [21] developed a spatial–spectral adaptive haze removal method for visible remote sensing images, which improves data quality from both spatial and spectral dimensions. Singh et al. [22] proposed an evolving visibility restoration model, employing dynamic differential evolution for the first time to optimize the model's control parameters. Meanwhile, the popular low-rank representation [23], minimal-maximal values prior [24], and bright channel prior [25] also provide new insights into remote sensing image dehazing. Physical model-based methods produce acceptable defogging results, but hand-designed prior knowledge cannot reflect all haze pollution characteristics,

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so the above approaches are only applicable to haze removal in specific scenarios.

Recently, deep learning techniques advanced significantly in a variety of computer vision applications, from image classification [26], [27] to semantic segmentation [28], [29]. Attributed to the powerful feature extraction and expression capabilities, many deep neural networks have been employed for defogging. Li et al. [10] presented a semi-supervised dehazing network for unpaired images from different regions. Ren et al. [11] used two networks to accomplish single-image dehazing: the coarse-scale network is used to predict a holistic transmission map, and the fine-scale network is used to refine local details. Qin et al. [30] created a residual structure-based convolutional neural network (CNN), in which multiscale convolutions are employed to mine multiscale haze features. In [31], residual convolutional layers were combined with residual deconvolutional layers to construct a deep symmetrical concatenation network, increasing the employment of remote sensing images. To add physical constraints to the deep learning network, Chen et al. [32] enabled a patch map-based generative adversarial network. Jiang et al. [33] proposed a deep learning network for remote sensing images under nonuniform hazy weather. Wen et al. [34] developed an encoder-free multiaxis physics-aware fusion network, which exhibits both light-weighted characteristics and computational efficiency. In addition, as a hot spot in the computer vision field, numerous remote sensing image dehazing studies have also been conducted on the attention mechanism [35], [36], [37], [38]. For example, Guo et al. [35] added a channel attention module to the end-to-end dehazing network to tap into the channel correlation. In [37], a memory-oriented generative adversarial network was proposed, where a dual region discriminator and a multistage attentive-recurrent memory structure are applied to enhance the unpaired remote sensing image defogging performance.

The above methods are applicable to three-band or multispectral remote sensing images, while HSIs have much finer spectral resolution, resulting in direct application in HSI dehazing does not yield satisfactory results. Up to now, only four publications have been developed for HSI haze removal, two of which are model-based approaches and the other two are deep learning-based approaches. Kang et al. [39] constructed a fog model that requires manually selecting two pixels of the same object to estimate haze abundance, where one pixel is damaged by haze and the other is clean. To automate the model, Li et al. [40] designed a saliency measure technique to determine the haze abundance of spectral channels automatically. These two HSI defogging methods are simple and efficient. However, restricted by the principle, the HSIs that can be processed must contain visible bands and infrared bands, which restricts the application scope of the methods. Moreover, Ma et al. [41] proposed a deep learning-based defogging network that can acquire spatial–spectral features from foggy and fog-free image pairs. Xu et al. [42] presented an asymmetric attention network, which employs multipath connections to capture haze features. In practice, acquiring paired hyperspectral data is costly in time and money, so the application performance is limited.

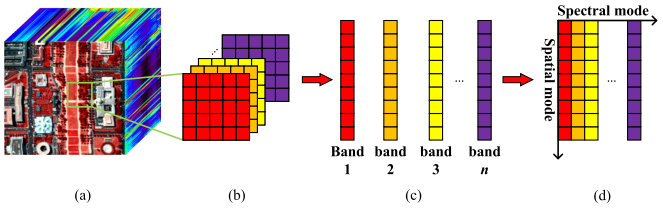


Fig. 1. Low-rank matrix representation of a clean HSI patch. (a) An original HSI cube. (b) A subcube. (c) Vectors. (d) A 2-D matrix.

Both model-based and deep learning-based approaches possess unique advantages and disadvantages. How to integrate the hybrid advantages of the two and establish an algorithm that synergizes the model with the deep neural network is still a challenging task.

B. Low-Rank Prior of HSI

HSIs have strong correlations between spectral bands, implying the presence of low-rank prior structures in clean HSIs. Fig. 1 illustrates the matrix representation of a clean HSI patch. According to the linear spectral mixture model [43], the hyperspectral data can be decomposed to the product of an endmember matrix and an abundance matrix, which is formulated as

$$\mathbf{X} = \mathbf{UV} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{HW \times B}$ is the hyperspectral data, in which H is the width of HSI, W is the height of HSI, and B is the number of spectral bands. $\mathbf{U} \in \mathbb{R}^{HW \times r}$ is the abundance matrix, where each row represents the linear combination coefficient of a certain pixel. $\mathbf{V} \in \mathbb{R}^{r \times B}$ is the endmember matrix, where each row stands for an endmember vector. r is the number of endmembers, and $r \ll B$. Due to the transitivity of low-rank matrices: $\text{rank}(\mathbf{X}) \leq \min(\text{rank}(\mathbf{U}), \text{rank}(\mathbf{V})) = r$, which suggests the 2-D matrix $\mathbf{X}$ is low-rank.

C. Deep Image Prior

The DIP framework [44] shows that CNNs can perform image restoration without training data, using only one degraded image. The neural network itself is adequate for learning the inherent image properties. Specifically, the network randomly initializes the input and adjusts the network weight by calculating the difference between the output image and the degraded image, thereby gradually optimizing the quality of the output image. The optimization process can be described as

$$\begin{aligned} \Theta^* &= \arg\min_{\Theta} \|\mathcal{Y} - f_\Theta(\mathcal{Z})\|_F^2 \\ \mathcal{X} &= f_{\Theta^*}(\mathcal{Z}) \end{aligned} \tag{2}$$

where f is the CNN, Θ is the network parameter, $\mathcal{Z} \in \mathbb{R}^{H \times W \times B}$ is the random initialization input, $\mathcal{Y} \in \mathbb{R}^{H \times W \times B}$ is the degraded image, and $\mathcal{X} \in \mathbb{R}^{H \times W \times B}$ is the restored image.

As the gradient descent algorithm continues optimizing the network parameters, the CNN will first fit the signal part, then fit the noise part (overfitting), and the final output $f_\Theta(\mathcal{Z})$

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will converge to degraded image $\mathcal{Y}$. Therefore, we need to set reasonable stopping conditions to terminate the iterative process before the network overfits. At this time, the optimal recovery result $\mathcal{X} = f_{\Theta^*}(\mathcal{Z})$ is obtained, where Θ^* stands for the network parameters at the appropriate moment.

III. Methodology

Fig. 2 exhibits the network architecture of the proposed HPC-UDN, consisting of four modules: 1) low-rank generation module; 2) spectral grouping module; 3) SGFF module; and 4) residual block (RB). Specifically, the desired 3-D hyperspectral data are obtained by low-rank matrix decomposition, where the decomposition factors are learned through two generative networks. Then, the hyperspectral data is partitioned into multiple spectral groups according to spectral correlation. An SGFF module is constructed to transfer spectral information and spatial features between two adjacent spectral groups. In addition, we employ a global skip connection (SC) to preserve more details and semantic information in the original data. Details of the four modules are shown below.

A. Low-Rank Generation Module

Relying on the low-rank property, many approaches have been proposed for defogging RGB or three-band remote sensing images [23], [45]. However, these methods usually require complex optimization techniques, such as the alternating direction method of multipliers (ADMMs) [46] algorithm or the augmented Lagrange multiplier (ALM) [47], leading to high computational complexity. To leverage the low-rank property while reducing the computational burden, we decide to combine deep networks with low-rank decomposition to tackle the haze removal problem in HSIs. Specifically, we use the U-shaped network architecture to build two generative networks. As shown in Fig. 3, the Unet2D-Spatial network and Unet1D-spectral network are applied to generate the left eigenvector matrix and right eigenvector matrix of matrix decomposition, respectively. The former aims to capture spatial information, and the latter aims to model spectral correlations. Both deep generative networks use randomly initialized inputs, and as the iteration progresses, the model parameters are continuously updated through the stochastic gradient descent algorithm.

We choose to employ two deep generative networks for low-rank matrix decomposition for three main reasons. First, low-rank matrix decomposition separates hazy and clean hyperspectral data, which makes it easier to remove haze and reduce redundant information. Second, the inherent inductive bias of deep neural networks enables the generation of implicit regularization signals, aiding in preventing overfitting and enhancing model generalization. Finally, the nonlinear modeling capabilities of deep neural networks can better capture complex relationships and structures in HSI, which is beneficial in restoring details and information lost in hazy images. In summary, the proposed low-rank generation module combines the simplicity of low-rank matrix decomposition with the powerful feature extraction capabilities of deep generative networks, significantly improving the performance of dehazing.

B. Spectral Grouping Module

It is universally acknowledged that HSIs contain redundant and highly correlated spectral information. Treating all the bands as a whole without considering their interactions will lead to an excessive number of parameters, which will reduce the computational efficiency. Hang et al. [48] find the spectral dimension grouping processing strategy can effectively utilize the rich information in HSIs. Utilizing multiple bands for better processing is similar to multimodal information processing discussed in many studies, such as audio-visual signal processing [49]. Therefore, we introduce the spectral grouping module [41] to partition the 3-D hyperspectral data into several spectral groups according to the CCs between spectral bands, calculated as

$$\mathrm{CC}_{i,j} = \frac{\mathrm{Cov}(b_i, b_j)}{\sqrt{\mathrm{Var}(b_i)\mathrm{Var}(b_j)}} \tag{3}$$

where $\mathrm{CC}_{i,j}$ is the CC between the ith and jth bands, b_i is the ith band, and $\mathrm{Cov}(\cdot)$ and $\mathrm{Var}(\cdot)$ are the covariance and variance, respectively.

Partitioning HSIs into multiple spectral groups based on CCs brings several significant benefits to hyperspectral haze removal. First, this spectral grouping approach enhances feature utilization. Grouping more similar and complementary bands into the same subgroup helps capture more comprehensive spectral features, thereby improving the accuracy of the dehazing result. Second, since this spectral grouping approach is based on CCs, it can adapt to HSIs with varying wavelength ranges and numbers of bands, enhancing the method's flexibility and applicability to various HSI processing tasks. Finally, this spectral grouping approach boosts the adaptability of the dehazing algorithm by applying different filter weights according to the varying haze levels within different spectral groups, increasing the ability to handle the diverse haze conditions presented in HSIs. In summary, partitioning spectral groups based on correlation provides multiple advantages for HSI dehazing, better meeting the practical application requirements.

C. SGFF Module

Available studies have shown that the longer wavelength bands can provide spectral information for the shorter wavelength bands [50]. Inspired by this, we designed an SGFF module to gradually exchange and fuse spectral features from long to short wavelength bands. As shown in Fig. 4, for two neighboring spectral groups, we first extract features in parallel using three convolutional layers of different sizes, after which we concatenate the three feature maps. Second, after applying element-wise summation to aggregate the multiscale features from the two spectral groups, we employ the global average pooling (GAP) to squeeze the global spatial information into a channel descriptor with a dimension of $1 \times 1 \times C$. To motivate the feature map, we first adopt a 1×1 convolutional layer to decrease the feature dimension (the reduction ratio is r), followed by a rectified linear unit (ReLU) activation function. Then, a 1×1 convolutional layer is exploited in parallel to restore the feature dimension to C dimensions, followed by a softmax activation function.

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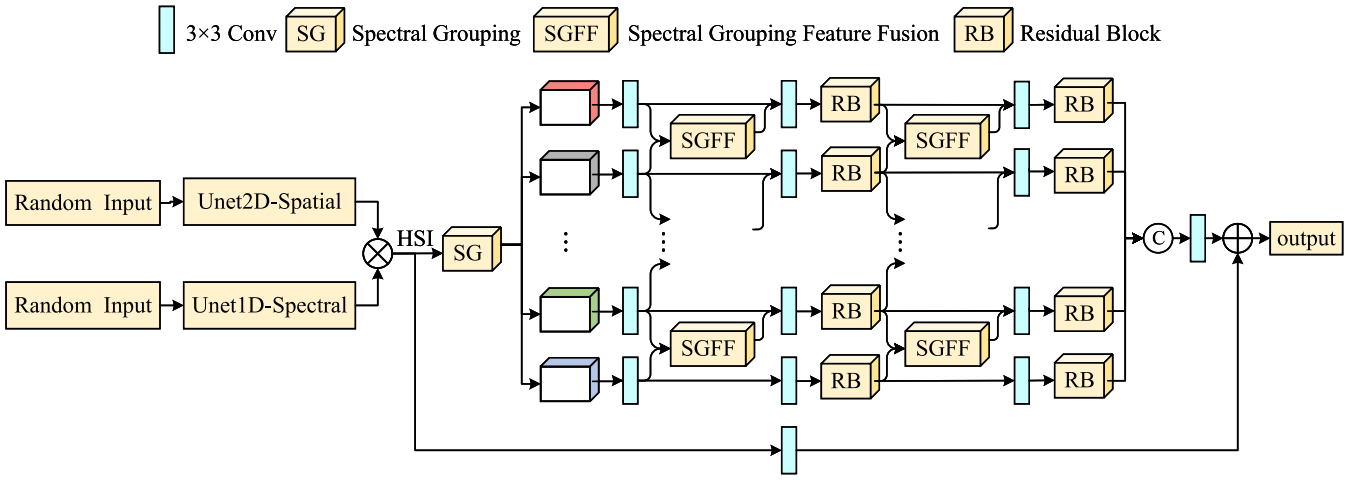


Fig. 2. Network architecture of the HPC-UDN.

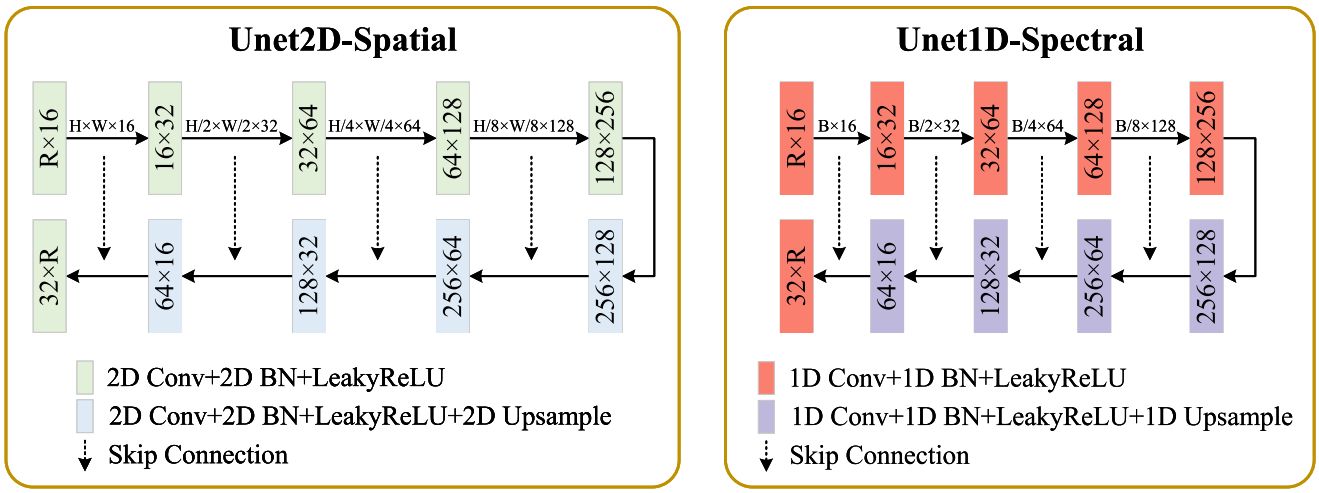


Fig. 3. Architectures of the deep low-rank generative networks.

Finally, we adaptively recalibrate multiscale features to better understand and utilize the characteristics of the input data.

By utilizing the self-attention mechanism to fuse spatial–spectral information of adjacent spectral groups, the SGFF module can further mine the correlation and complementarity between spectral groups. This information fusion approach enables the SGFF module to better transfer spatial and spectral information from longer-wavelength bands to shorter-wavelength bands, significantly enhancing the defogging performance. Through this operation, the SGFF module effectively eliminates the interference of haze on HSI, not only improving image clarity but also preserving more details and color information during the restoration process. Even in dense fog conditions, the proposed method can still generate vibrant and high-clarity images. It is significant to mention that the basic role of the SGFF module is information fusion, making it highly versatile and adaptable to various application scenarios.

D. Residual Block

Since ground objects typically vary in size across nonlocal regions in remote sensing images, contextual information may have an impact on multiscale feature representation [51]. To expand the receptive field while reducing the model parameters, we adopt a multiscale processing approach. As shown in Fig. 5, we construct hierarchical residue-like connections within a single RB to represent multiscale features at the granularity level. After the input feature map undergoes a 1×1 convolution, the feature map is divided into three routes. The first route $\mathbf{X}_1$ is not processed and directly transmitted to $\mathbf{Y}_1$; the second route $\mathbf{X}_2$ is divided into two branches after a 3×3 convolution, in which one continues to be propagated forward to $\mathbf{Y}_2$ and the other to $\mathbf{X}_3$ so that the third route obtains the information of the second route. By combining smaller filters, the equivalent receptive field is increased. Finally, the three feature maps are contacted and transmitted to a 1×1 filter to fully integrate multiscale features. In addition, the residual connection path applies a 1×1 convolution.

In the RB, the design of the extended receptive field enables the network to comprehensively describe the texture details of shallow features and the semantic clues of deep features, which improves the dehazing effect while preserving the realism and clarity of HSI. On the other hand, the multiscale feature representation enhances the generalization ability of the network, allowing the method to achieve

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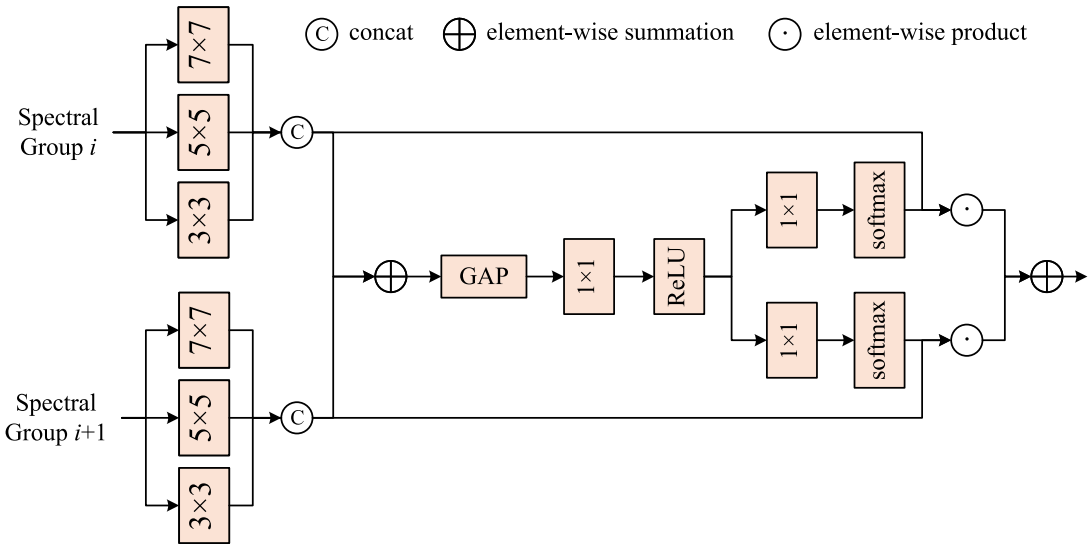


Fig. 4. Structure of SGFF module.

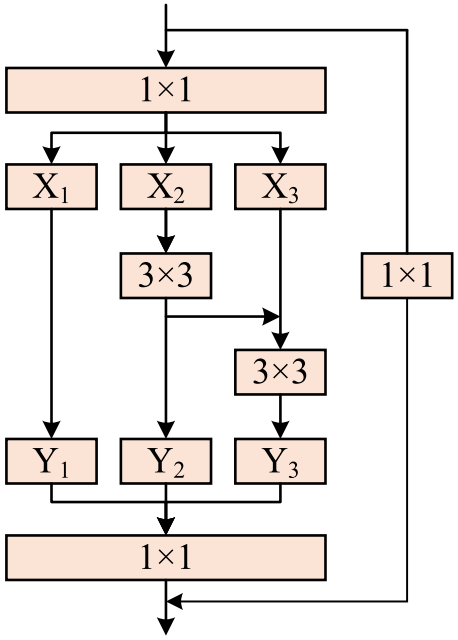


Fig. 5. Structure of RB.

better dehazing results for different input images and haze conditions.

E. Loss Function

We use mean absolute error (MAE) loss to optimize the parameters of the proposed HPC-UDN, which is calculated as

$$\mathrm{MAE}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{1}{N}\sum_{i=1}^{N}|\mathbf{Y}_i - \hat{\mathbf{Y}}_i| \tag{4}$$

where $\mathbf{Y}$ is the HSI after dehazing, $\hat{\mathbf{Y}}$ is the original hazy HSI, and N is the number of spectral bands.

The main purpose of choosing a simple loss function is to emphasize the regularization ability of deep CNN. We need to emphasize that the proposed algorithm uses no additional labeled data in its iterative process. It can autonomously mine the unknown prior information from the hazy data by simply utilizing the original hazy image for unsupervised learning. This unsupervised learning property endows the proposed algorithm with a high degree of flexibility and generality. In many cases, acquiring large amounts of labeled hyperspectral data is very difficult, and even nearly impossible in some specific contexts. Unsupervised learning can get rid of the dependence on labeled data and learn useful features directly from the original data, thus greatly expanding the application scenarios of the algorithm.

IV. Experiments

Extensive simulated and real-data experiments are conducted to examine the dehazing effect of the proposed approach. The results are contrasted with two dehazing techniques in the computer vision field: DCP [52] and SLP [53], three dehazing techniques in the remote sensing field: DHIM [8], ADM [9], and RLDP [54], and a dehazing technique in the HSI field: Defog [39].

A. Datasets

The experiments make use of the following three hyperspectral datasets, with the gray scale values in each band having been individually standardized. In our experiments, the size of the simulated hazy image is 256 × 256, and the size of the real-data hazy image is 512 × 512.

1) AVIRIS: The AVIRIS datasets[1] are collected by the Airborne Visible InfraRed Imaging Spectrometer and include 224 spectral bands in the range of 0.4–2.5 μm. The spatial resolution of these images is about 15 m. We crop six images for experiments.

[1]https://aviris.jpl.nasa.gov/data/index.html

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2) GF-5: The GF-5 datasets[2] are collected by the Advanced Hyperspectral Imager of the GF-5 satellite. This sensor records 330 continuous spectral bands in the range of 0.4–2.5 μm. The spatial resolution of these images is 30 m/pixel; 25 bands located at No. 193–200 and 246–262 were removed from the dataset due to water and atmospheric absorption. We crop six images for experiments.

3) Chikusei: The Chikusei datasets[3] are collected by Headwall's Hyperspectral Visible and Near-Infrared, series C imaging sensor and include 128 spectral bands in the range of 0.363–1.018 μm. The spatial resolution of these images is 2.5 m. We crop ten images for experiments.

B. Parameter Setting

We use the Adam optimizer [55] to update network parameters, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 1e-8$. We initialize the learning rate to $1e-3$. The iteration epoch varies from 2000 to 4000. In the low-rank generation module, the rank-8 ($R = 8$) model is used for the AVIRIS dataset, the rank-10 ($R = 10$) model is used for the GF-5 dataset, and the rank-4 ($R = 4$) model is used for the Chikusei dataset. By calculating the CC, the AVIRIS dataset is divided into seven spectral groups, the GF-5 dataset is divided into five spectral groups, and the Chikusei dataset is divided into two spectral groups. In the SGFF module, the reduction ratio r is set to 8. The entire model is built based on the Pytorch environment and is iteratively optimized using a NVIDIA 3090 GPU.

C. Simulated-Data Experiments

To more realistically model fog pollution, we adopt the haze synthesis method of [35]. Specifically, the fog mask is extracted from the cirrus band of Landsat-8 OLI and then superimposed on the original image according to the wavelength ratio. This haze synthesis method has two advantages. On the one hand, the spatially nonuniform properties of actual haze can be represented by the cirrus band's reflectance. On the other hand, it also considers how each band's pollution level is affected by wavelength. That is, the amount of haze gradually reduces as wavelength increases. The model for haze imaging is given as

$$\mathbf{I}_i = \mathbf{J}_i e^{\left(\frac{\lambda_1}{\lambda_i}\right)^{\gamma} \ln \mathbf{t}_1} + A\left(1 - e^{\left(\frac{\lambda_1}{\lambda_i}\right)^{\gamma} \ln \mathbf{t}_1}\right) \tag{5}$$

where $\mathbf{I}$ is the hazy HSI, $\mathbf{J}$ is the clean HSI, A is the global AL, λ is the wavelength, and γ is the spatial function, which is set to 1. $\mathbf{t}_1$ is the reference transmission map, calculated from the reflectance of the cirrus band

$$\mathbf{t}_1 = 1 - \omega \mathbf{P}_9 \tag{6}$$

where ω is the weighting factor that determines haze level, and $\mathbf{P}_9$ is the reflectance of the cirrus band. According to (6), the larger the weighting factor ω is, which indicates the heavier the haze condition, the smaller calculated reference transmission map $\mathbf{t}_1$ will be. In our experiment, we set up five intensities of haze, i.e., $\omega = [0.2, 0.4, 0.6, 0.8, 1]$.

[2]https://grid.cpeos.org.cn/home.htm
[3]https://naotoyokoya.com/Download.html

TABLE I
QUANTITATIVE ASSESSMENT ON AVIRIS DATASET

Index	Method	0.2	0.4	0.6	0.8	1
CC ↑	DCP	0.9655	0.9195	0.8760	0.8274	0.7675
	DHIM	0.9694	0.9253	0.8805	0.8314	0.7721
	ADM	0.9725	0.9312	0.8824	0.8368	0.7957
	SLP	0.9761	0.9399	0.8902	0.8494	0.8041
	RLDP	0.9809	0.9444	0.9020	0.8618	0.8135
	Defog	0.7849	0.8191	0.7941	0.8419	0.8308
	Ours	**0.9881**	**0.9632**	**0.9291**	**0.8892**	**0.8313**
UIQI ↑	DCP	0.9433	0.8906	0.8415	0.7877	0.7267
	DHIM	0.9538	0.9016	0.8534	0.7997	0.7387
	ADM	0.9596	0.9139	0.8623	0.8117	0.7532
	SLP	0.9617	0.9170	0.8731	0.8309	0.7796
	RLDP	0.9725	0.9234	0.8827	0.8375	0.7847
	Defog	0.6660	0.6620	0.7035	0.7746	0.7743
	Ours	**0.9788**	**0.9339**	**0.9006**	**0.8551**	**0.8047**
RMSE ↓	DCP	0.0166	0.0342	0.0536	0.0772	0.0984
	DHIM	0.0058	0.0189	0.0365	0.0593	0.0814
	ADM	0.0040	0.0108	0.0303	0.0541	0.0696
	SLP	0.0042	0.0115	0.0297	0.0523	0.0733
	RLDP	0.0025	0.0084	0.0208	0.0370	0.0568
	Defog	0.0864	0.0925	0.0864	0.0555	0.0529
	Ours	**0.0014**	**0.0080**	**0.0194**	**0.0342**	**0.0481**
SAM ↓	DCP	2.5825	4.9902	7.3545	9.6421	11.8521
	DHIM	1.0750	3.2025	5.7221	8.2963	10.8255
	ADM	0.6753	2.4054	4.9788	7.5265	9.8707
	SLP	0.5978	2.7633	5.2929	7.8061	10.6188
	RLDP	0.4701	1.8773	3.1924	4.7168	6.4738
	Defog	8.0230	10.2648	9.7808	8.6217	8.7592
	Ours	**0.3176**	**1.0850**	**2.4562**	**4.0569**	**5.9565**
MPSNR ↑	DCP	20.7465	19.9781	19.0739	18.4265	17.6810
	DHIM	24.1428	23.1001	21.9003	20.5872	19.0777
	ADM	30.3603	29.2414	27.6597	24.9552	22.1723
	SLP	36.5824	32.0219	28.1969	25.8712	22.3220
	RLDP	37.8988	32.2018	29.0763	26.2935	23.3726
	Defog	31.4202	31.5199	28.4666	26.3044	24.8763
	Ours	**42.9199**	**34.5381**	**30.3974**	**29.5706**	**28.2769**
DEHAZEfr ↑	DCP	0.9410	0.9243	0.9156	0.8783	0.8463
	DHIM	0.9507	0.9451	0.9286	0.8948	0.8478
	ADM	0.9854	0.9683	0.9307	0.8955	0.8563
	SLP	0.9902	0.9713	0.9330	0.9008	0.8582
	RLDP	0.9943	0.9775	0.9467	0.9085	0.8590
	Defog	0.6215	0.6510	0.7098	0.7419	0.7258
	Ours	**0.9983**	**0.9873**	**0.9494**	**0.9092**	**0.8607**

In order to quantitatively evaluate the haze removal effect of seven methods, five widely employed metrics are adopted: CC, universal image quality index (UIQI), root mean square error (RMSE), spectral angle mapping (SAM), and mean peak signal-to-noise ratio (MPSNR). In addition, Min et al. [56] proposed some generalized dehazing quality evaluation methods for performance testing. In order to increase the breadth of the experiment, we also use the DEHAZEfr to conduct dehazing performance testing. Among them, the larger the values of CC, UIQI, MPSNR, and DEHAZEfr, the higher the clarity of the dehazing image. The smaller the values of RMSE and SAM, the higher the fidelity of the dehazing image.

1) Haze Removal Results: Tables I–III are the quantitative comparison results on the AVIRIS dataset, GF-5 dataset, and Chikusei dataset, respectively. The best metrics are bolded, and the next best are underlined. It is worth noting that due to the limitation of the principle, the images that can be processed by the Defog method [39] must contain visible

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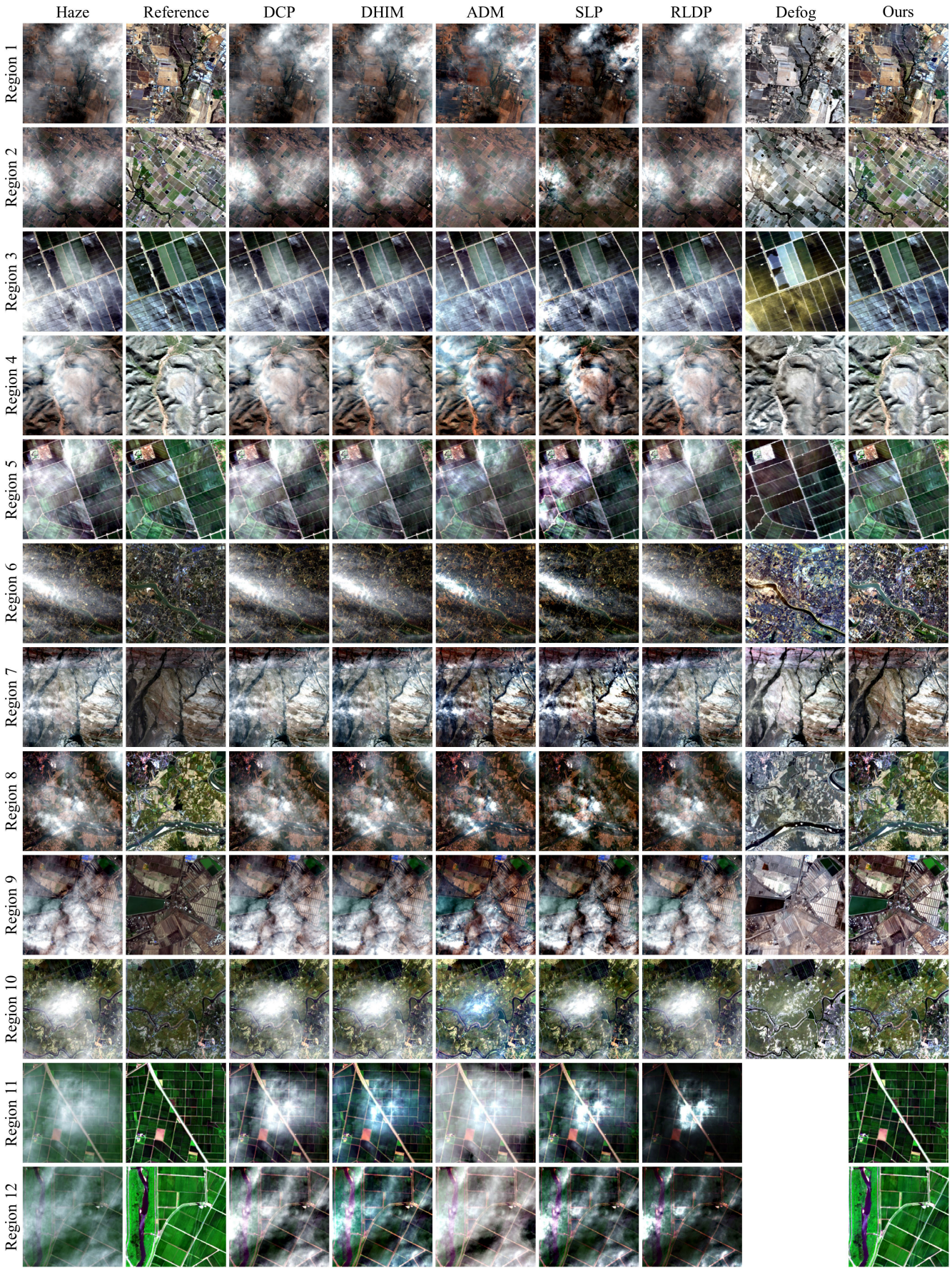


Fig. 6. Defogging results on the AVIRIS dataset (Regions 1–5), GF-5 dataset (Regions 6–10), and Chikusei dataset (Regions 11 and 12).

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TABLE II
QUANTITATIVE ASSESSMENT ON GF-5 DATASET

Index	Method	0.2	0.4	0.6	0.8	1
CC ↑	DCP	0.9827	0.9414	0.8886	0.8267	0.7509
	DHIM	0.9843	0.9439	0.8905	0.8281	0.7525
	ADM	0.9889	0.9515	0.8976	0.8453	0.7805
	SLP	0.9910	0.9518	0.9031	0.8470	0.7919
	RLDP	0.9925	0.9521	0.9042	0.8483	0.8037
	Defog	0.8134	0.8666	0.8943	0.8880	0.8640
	Ours	**0.9952**	**0.9789**	**0.9517**	**0.9121**	**0.8717**
UIQI ↑	DCP	0.9742	0.9222	0.8579	0.7859	0.7028
	DHIM	0.9829	0.9312	0.8693	0.7982	0.7142
	ADM	0.9856	0.9490	0.8791	0.8108	0.7307
	SLP	0.9927	0.9522	0.8893	0.8112	0.7389
	RLDP	0.9944	0.9556	0.8953	0.8314	0.7529
	Defog	0.7587	0.8204	0.8534	0.8482	0.8072
	Ours	**0.9960**	**0.9715**	**0.9347**	**0.8772**	**0.8234**
RMSE ↓	DCP	0.0242	0.0418	0.0623	0.0866	0.1150
	DHIM	0.0085	0.0232	0.0426	0.0668	0.0966
	ADM	0.0058	0.0133	0.0353	0.0599	0.0829
	SLP	0.0050	0.0124	0.0275	0.0451	0.0659
	RLDP	0.0044	0.0111	0.0265	0.0425	0.0620
	Defog	0.0999	0.0528	0.0478	0.0503	0.0556
	Ours	**0.0029**	**0.0110**	**0.0173**	**0.0275**	**0.0383**
SAM ↓	DCP	3.7554	6.2157	8.5433	10.6446	12.5268
	DHIM	1.5633	4.0342	6.6471	9.1589	11.4417
	ADM	0.9820	3.0301	5.7835	8.3090	10.4325
	SLP	0.8579	2.6591	4.0731	5.8862	7.4709
	RLDP	0.8350	2.2091	4.0576	5.5988	6.8850
	Defog	8.6266	5.7609	5.7856	5.5098	5.8221
	Ours	**0.5871**	**1.4864**	**2.4776**	**3.7375**	**4.9282**
MPSNR ↑	DCP	22.8232	21.3891	20.8181	19.8725	17.6975
	DHIM	25.0152	23.3820	21.8822	20.1987	18.4617
	ADM	30.1660	27.9830	25.9865	23.3510	20.6583
	SLP	32.2976	29.5474	26.1749	24.0328	22.1320
	RLDP	34.5656	31.7267	28.6728	26.1560	23.6366
	Defog	32.7273	30.5865	28.2259	26.3173	24.4286
	Ours	**38.3279**	**32.3571**	**29.7547**	**27.1915**	**25.5809**
DEHAZEfr ↑	DCP	0.9339	0.9198	0.8761	0.8494	0.8185
	DHIM	0.9576	0.9391	0.9064	0.8671	0.8266
	ADM	0.9590	0.9400	0.9070	0.8725	0.8273
	SLP	0.9661	0.9484	0.9133	0.8734	0.8286
	RLDP	0.9719	0.9510	0.9147	0.8784	0.8385
	Defog	0.6929	0.7504	0.7885	0.8043	0.8014
	Ours	**0.9773**	**0.9632**	**0.9175**	**0.8825**	**0.8395**

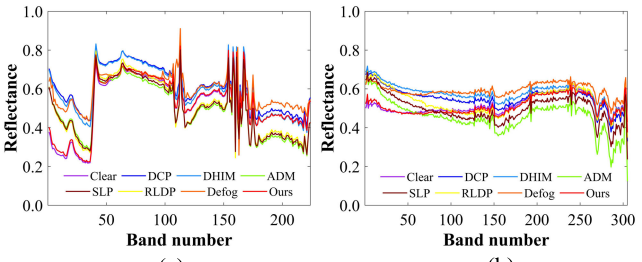


Fig. 7. Spectral reflectance of land covers. (a) Vegetation of AVIRIS dataset (122, 124). (b) Rock of GF-5 dataset (160, 109).

(0.36–0.65 μm) and near-infrared (2.01–2.36 μm) wavelength bands, so it cannot defog the Chikusei dataset. We can see that the proposed HPC-UDN outperforms other haze removal techniques on all three datasets, reflecting the robustness of the method. Various test indicators prove that the proposed

TABLE III
QUANTITATIVE ASSESSMENT ON CHIKUSEI DATASET

Index	Method	0.2	0.4	0.6	0.8	1
CC ↑	DCP	0.8820	0.8111	0.7366	0.6516	0.5104
	DHIM	0.9007	0.8427	0.7675	0.6562	0.5791
	ADM	0.9438	0.8877	0.8071	0.7136	0.6120
	SLP	0.9699	0.8968	0.8176	0.7470	0.6552
	RLDP	0.9861	0.9464	0.8824	0.7997	0.7190
	Ours	**0.9967**	**0.9679**	**0.9149**	**0.8391**	**0.7592**
UIQI ↑	DCP	0.7242	0.6242	0.5752	0.4595	0.3242
	DHIM	0.8219	0.7355	0.6092	0.5248	0.4656
	ADM	0.8372	0.7655	0.6829	0.6028	0.5313
	SLP	0.9562	0.8520	0.7358	0.6269	0.5419
	RLDP	0.9798	0.9193	0.8372	0.7407	0.6489
	Ours	**0.9952**	**0.9521**	**0.8789**	**0.7866**	**0.6933**
RMSE ↓	DCP	0.1388	0.2024	0.2037	0.2507	0.4046
	DHIM	0.0900	0.1218	0.1716	0.2304	0.2601
	ADM	0.0813	0.1000	0.1267	0.1531	0.1838
	SLP	0.0283	0.0562	0.0909	0.1314	0.1695
	RLDP	0.0202	0.0390	0.0605	0.0851	0.1171
	Ours	**0.0105**	**0.0308**	**0.0498**	**0.0746**	**0.1004**
SAM ↓	DCP	7.6912	9.3844	13.7645	17.7766	20.9749
	DHIM	6.6387	9.1227	11.3066	14.1066	16.6036
	ADM	5.7842	7.4168	9.8966	12.9808	16.5496
	SLP	2.8111	6.0209	9.4553	12.7935	16.0202
	RLDP	2.2798	5.2048	8.3227	11.3473	14.2663
	Ours	**1.1882**	**3.7205**	**6.7067**	**9.8712**	**13.0050**
MPSNR ↑	DCP	19.1830	16.6552	15.6944	12.5344	9.0658
	DHIM	22.3211	18.8840	16.5124	14.6453	13.8871
	ADM	22.6492	20.4193	18.4820	17.3863	16.2261
	SLP	31.7045	26.1449	21.9815	18.9125	16.8077
	RLDP	34.6276	29.3904	25.6036	22.9080	20.9196
	Ours	**40.1291**	**30.9679**	**27.0571**	**23.6601**	**21.8852**
DEHAZEfr ↑	DCP	0.7213	0.6681	0.6391	0.5883	0.4754
	DHIM	0.8012	0.7497	0.6971	0.6139	0.5546
	ADM	0.9157	0.8388	0.7396	0.6293	0.5602
	SLP	0.9440	0.8550	0.7579	0.6601	0.5766
	RLDP	0.9702	0.9185	0.8554	0.7799	0.7004
	Ours	**0.9887**	**0.9466**	**0.8888**	**0.8186**	**0.7393**

method effectively eliminates haze interference while retaining spectral information well.

Visual comparison results on simulated datasets are illustrated in Fig. 6, where Regions 1–5 are AVIRIS images (bands 29, 20, and 12), Regions 6–10 are GF-5 images (bands 59, 38, and 20), and Regions 11 and 12 are Chikusei images (bands 60, 40, and 20). To make the experiment more extensive, the simulated datasets contain a variety of land cover types, including buildings, farmland, rock, vegetation, and urban areas. It is evident that DCP, DHIM, ADM, SLP, and RLDP methods still contain a large amount of unremoved fog because these approaches are developed for three-band natural or remote sensing images, which makes them not applicable to HSI dehazing. In addition, due to the high brightness of the fog, the image contrast is poor. While the Defog method accomplishes a remarkable dehazing performance, the spectral distortion is severe. In contrast, the images provided by the proposed method have higher visibility and more accurate color fidelity.

To verify the dehazing effects of seven methods in the spectral dimension, Fig. 7 illustrates spectral reflectance curves of vegetation and rock, in which vertical axis stands for

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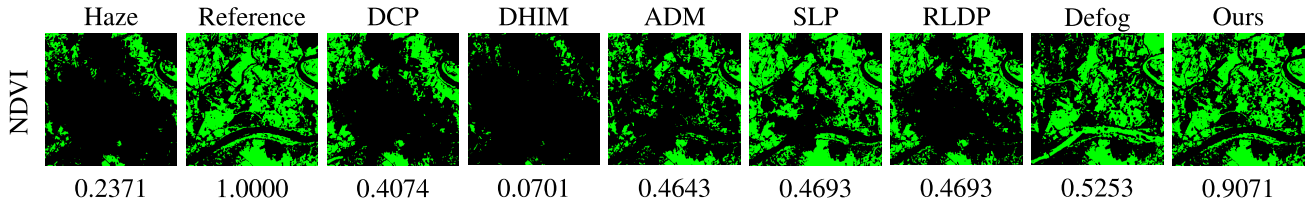


Fig. 8. Vegetation extraction results with NDVI (Region 8).

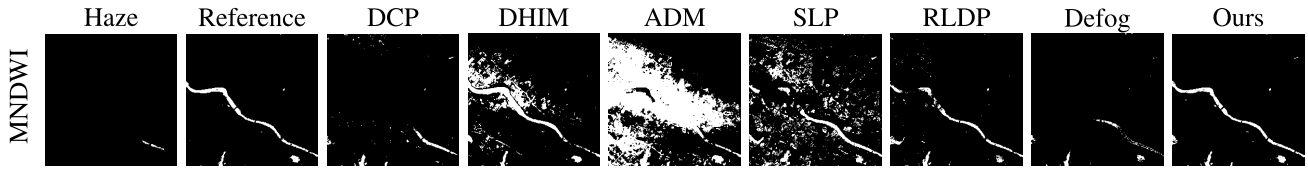


Fig. 9. Water body extraction results with MNDWI (Region 6).

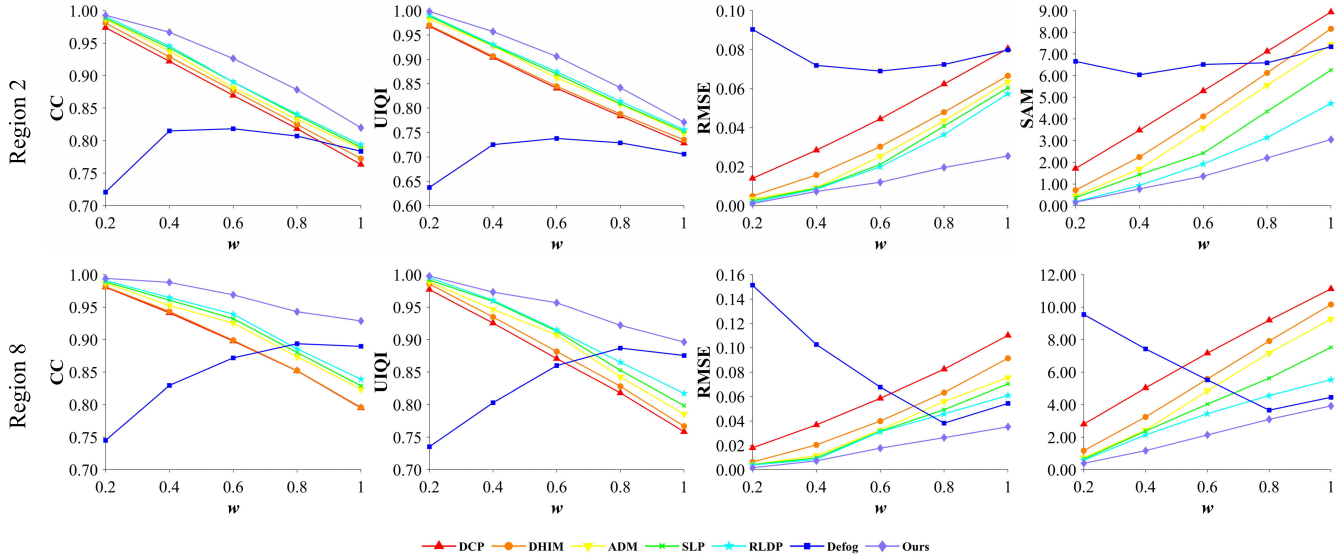


Fig. 10. Line graph of seven approaches in varying haze levels.

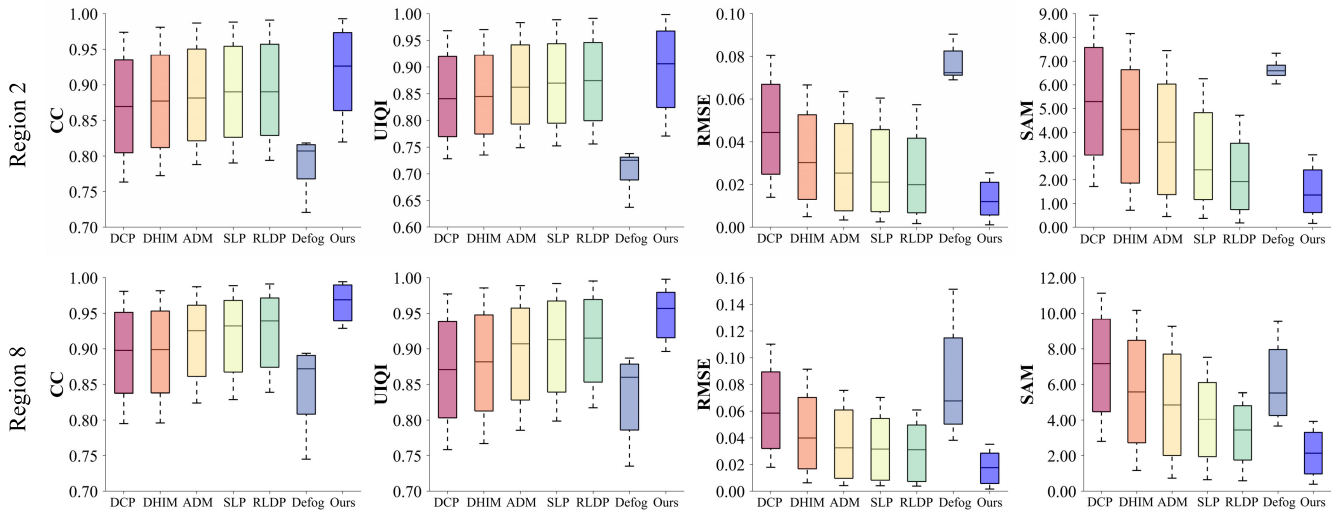


Fig. 11. Box-plot of seven approaches in varying haze levels.

normalized digital number (DN) and horizontal axis stands for spectral band. We can observe that the proposed HPC-UDN also has the best restoration effect, which is the closest to the original spectral curve. Experiments conducted both quantitatively and qualitatively demonstrate the superior dehazing effect of the proposed HPC-UDN and perform well in maintaining the spectral features of HSIs, which significantly improves the quality of the data.

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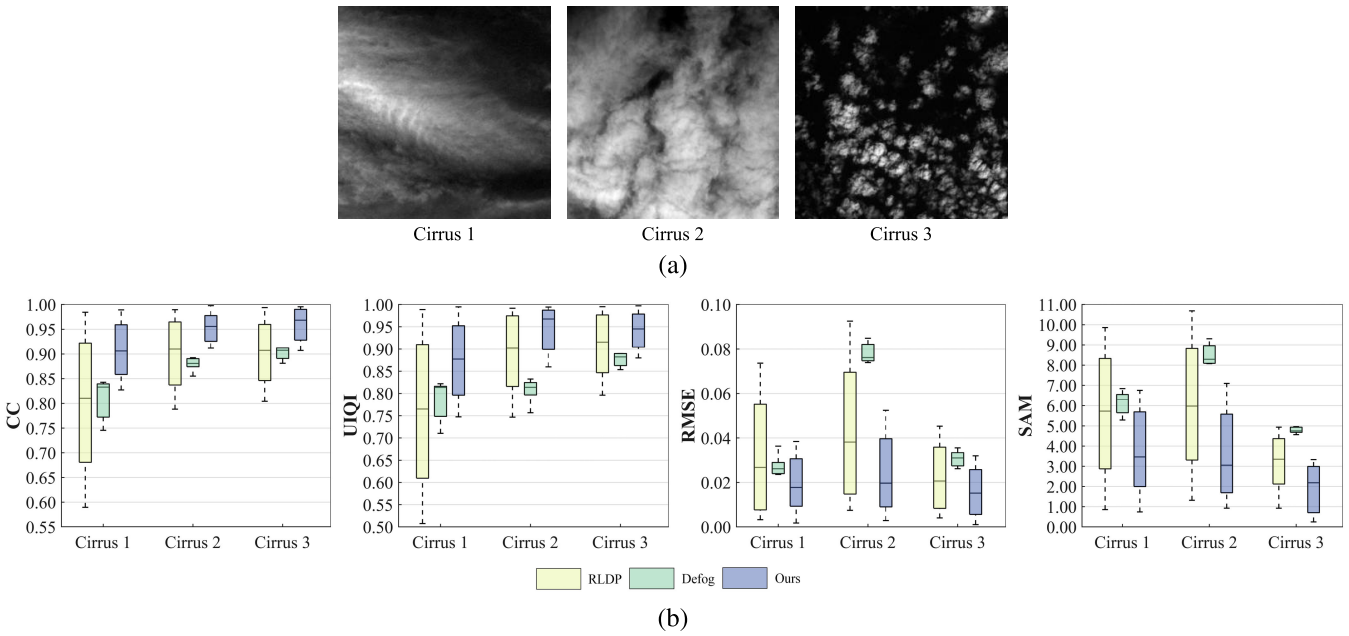


Fig. 12. Box-plot of three approaches in varying haze shapes. (a) Styles of fog masks. (b) Box-plot of three approaches in varying haze levels.

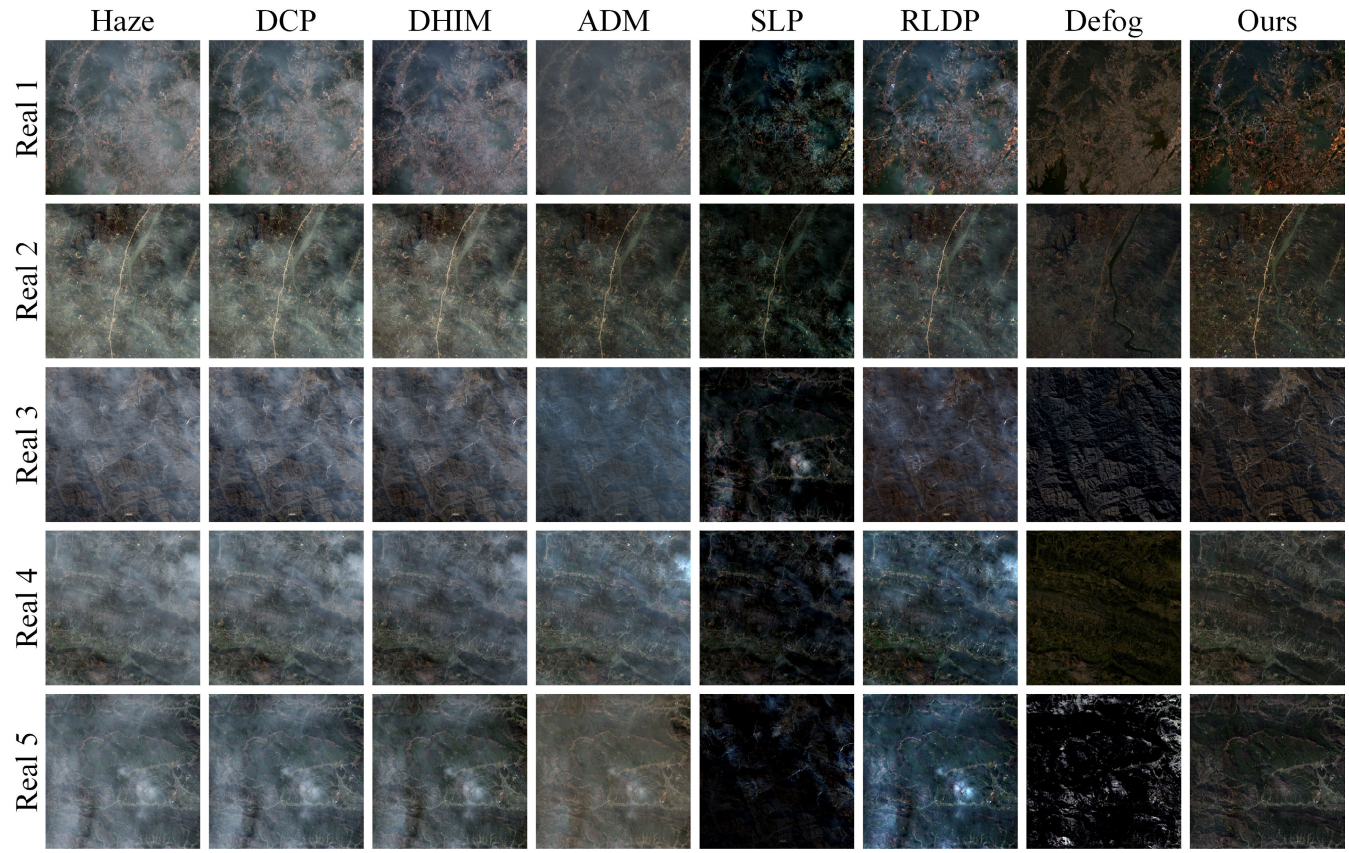


Fig. 13. Defogging results of real hazy GF-5 images.

2) Practical Application Analysis: To deeply check the restoration effect on the spectrum, we carry out two practical application experiments. Specifically, we employ the normalized difference vegetation index (NDVI) and the modified normalized difference water index (MNDWI) to extract vegetation and water bodies from dehazed images. NDVI and MNDWI are defined as follows, respectively,

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}} \tag{7}$$

$$\mathrm{MNDWI} = \frac{\rho_{\mathrm{Green}} - \rho_{\mathrm{MIR}}}{\rho_{\mathrm{Green}} + \rho_{\mathrm{MIR}}} \tag{8}$$

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where ρ_{NIR}, ρ_{Red}, ρ_{Green}, and ρ_{MIR} are the reflectance of NIR, red, green, and MIR band, respectively.

Vegetation extraction results for Region 8 are shown in Fig. 8. The threshold value of NDVI is 0.5, which means that areas with NDVI values greater than or equal to 0.5 are vegetation-covered areas (shown in green), and vice versa for nonvegetation-covered areas (shown in black). The number below the image is its IoU score. It is evident that when the HSI is disturbed by fog, the vegetation extraction is seriously affected, and the IoU score has a large deviation. After processing by the proposed method, the extraction result is more similar to the label.

Water body extraction results for Region 6 are shown in Fig. 9. The threshold value of water-land separation is 0.75, which indicates that areas with MNDWI values greater than or equal to 0.75 are water body areas (shown in white), and vice versa for land areas (shown in black). We can see that the DCP and Defog have excessive leakage for water body areas, while DHIM, ADM, and SLP mistakenly categorize many land areas as water body areas. In comparison, RLDP and the proposed method are more suitable for the water body extraction task.

3) Haze Level Analysis: To test the generalization performance for different haze levels, we carry out comparative experiments on Regions 2 and 8, the results are shown in Figs. 10 and 11. As the weighting factor ω progressively rises (the haze concentration gradually increases), it is evident the accuracy of each method decreases. However, the proposed HPC-UDN is more steady in accuracy and more flexible to varying haze levels than current dehazing techniques.

4) Haze Shape Analysis: In addition to the haze level affecting the haze removal capability, the haze shape also has a specific impact. Therefore, we extract three representative fog masks from the cirrus band of Landsat-8 OLI for experiments. The styles of the three fog masks are shown in Fig. 12(a), where Cirrus 1 simulates the thin fog case, Cirrus 2 simulates the thick fog case, and Cirrus 3 simulates the point-like fog case. As mentioned, we set 5 haze thicknesses ($\omega = [0.2, 0.4, 0.6, 0.8, 1]$). Fig. 12(b) displays the box-plot of RLDP, Defog, and proposed HPC-UDN. In three circumstances, the proposed HPC-UDN outperforms the other two approaches, proving its good robustness.

D. Real-Data Experiments

To examine the practical applicability of all haze removal methods, we collect five GF-5 hazy images (Real 1–5) for real-data experiments. Visual results (bands 59, 38, and 20) are shown in Fig. 13. Due to the large discrepancy between multispectral and hyperspectral data, the dehazed images provided by DCP, DHIM, ADM, SLP, and RLDP exhibit significant residual haze. Although the Defog method produces clear dehazed results, there is apparent spectral distortion (see the clean pixels in the original images). In contrast to existing approaches, the proposed HPC-UDN effectively removes the haze and successfully maintains the spectral information.

In addition, the water body extraction experiment is carried out for the dehazed images on Real 2, where the haze-free image is derived from Google Maps. Specifically, we extract RGB bands from images dehazed by all methods, and then apply the multiclass support vector machine (SVM) [57] to extract river regions. As shown in Fig. 14, the image processed by the proposed HPC-UDN has fewer misclassified pixels and can extract the river region more accurately.

TABLE IV
ABLATION STUDY OF DIFFERENT NETWORK CONSTRUCTION

	CC ↑	UIQI ↑	RMSE ↓	SAM ↓
without SC	0.9552	0.9384	0.0484	5.4508
without RB	0.9706	0.9678	0.0340	3.7059
without SGFF	0.9650	0.9441	0.0415	4.7387
Ours	**0.9849**	**0.9802**	**0.0255**	**3.4569**

TABLE V
COMPARISON OF DEHAZING RESULTS WITH DIFFERENT REDUCTION RATIO

	CC ↑	UIQI ↑	RMSE ↓	SAM ↓
2	0.9758	0.9712	0.0394	2.6059
4	0.9865	0.9881	0.0158	1.4971
8	**0.9890**	**0.9896**	**0.0137**	**1.1643**
16	0.9816	0.9783	0.0271	2.4144
32	0.9626	0.9542	0.0487	3.8438

E. Ablation Study

1) Effectiveness of Architectural Components: We carry out ablation experiments on Region 1 to study how architectural components affect the dehazing performance. As shown in Table IV, SC can accelerate convergence while improving the capacity to express features. Therefore, removing the SC will drastically impair the proposed HPC-UDN's dehazing effectiveness. In addition, the SGFF module performs information exchange between two adjacent spectral groups, which transfers spatial–spectral features from the longer wavelength group (less haze pollution) to the shorter wavelength group (more haze pollution), resulting in a performance improvement. Similarly, the RB has a positive impact on HSI haze removal. According to the ablation results, the HPC-UDN proposed in this work produces the best results.

2) Parameter Sensitivity of Rank: The selection of rank also significantly impacts the haze removal performance. We conducted comparative experiments on the Region 4 image of the AVIRIS dataset, Region 10 image of the GF-5 dataset, and Region 12 image of the Chikusei dataset with haze level $\omega = 0.6$. The results are shown in Fig. 15. We can observe that the rank-8 ($R = 8$) model is preferable for the AVIRIS dataset, the rank-10 ($R = 10$) model gets superior results on the GF-5 dataset, and rank-4 ($R = 4$) is suitable for the Chikusei dataset.

3) Parameter Sensitivity of Reduction Ratio: The reduction ratio is a hyperparameter that can affect the performance of the SGFF module. We conducted comparative experiments on the Region 4 image with haze level $\omega = 0.2$. As shown in Table V, when the reduction ratio r is set to 8, the dehazing performance of the proposed method is the best.

F. Discussion

A common problem for DIP-based unsupervised methods is to choose the optimal iteration count. As depicted in Fig. 16,

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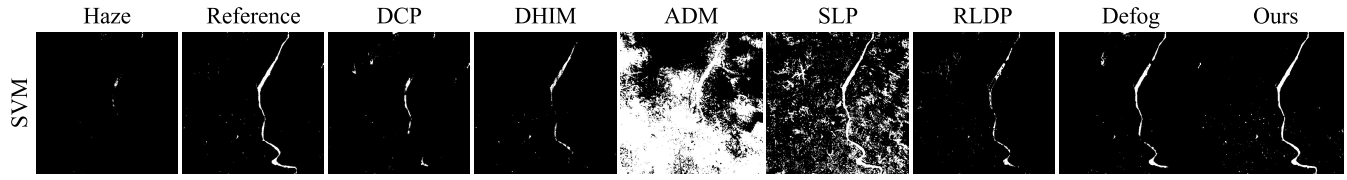


Fig. 14. Water body extraction results of real hazy GF-5 images.

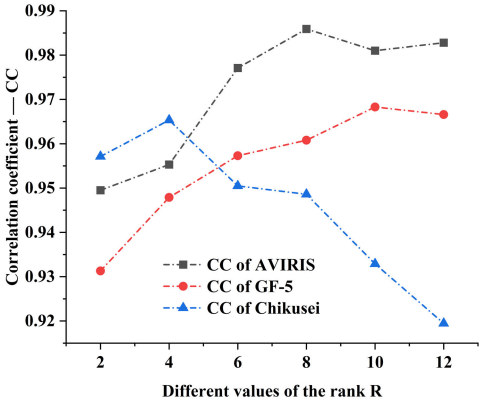


Fig. 15. Comparison of dehazing results on three datasets with different ranks.

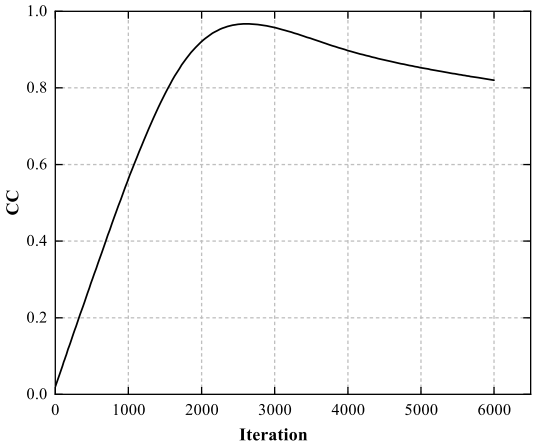


Fig. 16. CC of output image derived by proposed method at different iterations.

the approach tends to fit the clean image at the early stage of iterations, and the performance index gradually improves. It plateaus at about 2600 iterations and reaches the best performance index. After that, the network starts to misidentify some haze pollution as the recovered image and the performance metric gradually drop. However, there is limited reliable early stopping criterion. For the proposed method, we determine the optimal time to terminate the iterative process based on the visual quality assessment of the recovered hyperspectral data. Specifically, in simulation experiments, referring to the quantitative metrics of the recovered data can avoid overfitting and obtain the optimal model effect. For real experiments, we need to manually select the optimal image from the restored HSI results regularly saved by the model. The automatic stopping criterion for optimal approximation accuracy will be left to future research.

Another problem for DIP-based unsupervised methods is the balance between accuracy and time efficiency. In fact, the proposed HPC-UDN can reconstruct the HSI from a single hazing HSI as input. To improve the efficiency of the proposed HPC-UDN for large-scale HSI processing, several strategies, such as lightweight, knowledge distillation, and pruning strategies, can be utilized to improve the network. Furthermore, the big model can cooperate with the DIP-based model to achieve the learning of the comment features.

V. Conclusion

In order to integrate the complementary benefits of deep learning-based and physical model-based approaches, we have proposed an HPC-UDN for HSI haze removal, which effectively integrates low-rank prior, deep priors, and physical haze prior. First, instead of directly reconstructing the HSI data, we transform the HSI dehazing recovery problem to extrapolate the spectral basis and spatial representation coefficients via low-rank matrix factorization. Then, multiple spectral groups are divided in accordance with the spectral correlation and complementarity. Considering the relationship between wavelength and haze pollution level, we designed a novel SGFF module to transfer spectral and spatial information between adjacent spectral groups. Finally, each spectral branch extracts features in parallel to recover clean HSI. In addition, a global SC is employed to mitigate the model burden. Extensive experiments indicate the effectiveness and robustness of proposed HPC-UDN and potential applications in the GF-5 image dehazing task.

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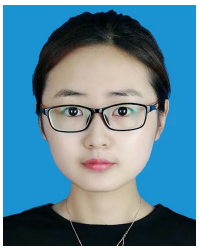
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